Solving the Duffing Equation Using Numerical Methods

PHYS 339: Final Project

May 4, 2020

1 Introduction

Imagine a man walking on a street on his way to work. There is a tiler working on the roof of a building on his street. If the man leaves his house at a certain time, a tile that the tiler drops from the roof will fall on his head and the man will die. If the man leaves his house a second later, the trajectory of his life would turn out to be very different [2].

This is an example that was presented by a French mathematician and physicist, Henri Poincare (1854-1912). In 1988, he published two papers on celestial mechanics. In these books, he presented the methods of evaluating nonlinear differential equations and contributed to the development of chaos theory [3]. Chaos theory describes a dynamical system that is extremely sensitive to its initial conditions [1]. Learning how these nonlinear situations evolve over time is important in understanding many real-life situations, such as chemical reactions, population growth, and even the spread of an infectious disease [4].

In this paper, we will examine a certain type of nonlinear system, the Duffing Oscillator, which is a nonlinear, second-order differential equation used to model damped, driven oscillators. Due to the fact that nonlinear systems are difficult to evaluate analytically, we used Python to assist us in numerically solving these complex differential equations and in visualizing the changes of this system.

2 Background

Earlier this semester, we evaluated simple harmonic motion and how the motion of a system can change with the introduction of forces, such as damping force and driving force. The type of motion we evaluated previously deal with linear motion, in which the equation of motion depends on a linear term, x.

In the last couple of decades, however, physicists have discovered ways to analyze nonlinear motion, in which the equation of motion depends on a higher-order x term. These nonlinear relations cause a phenomenon referred to as "chaos."

Poincare states that "chaotic" systems are extremely sensitive to initial conditions, which will drastically change its motion, making the trajectory "nonlinear" or "chaotic." The Duffing Oscillator has an x^3 term which prevents it from obeying Hooke's law and causes it to be sensitive to its initial conditions, potentially showing chaotic behavior.

3 The Solution

The equation of motion of a Duffing Oscillator is given by the expression,

$$m\ddot{x} + \beta\dot{x} + \alpha x + \gamma x^3 = Fcos(\omega t), \tag{1}$$

where m is mass, F is the magnitude of the driving force, and ω is the angular frequency of the movement of the force. The constants are defined as follows: $\beta = 0.05$, $\alpha = -3$, and $\gamma = 1$. Here, the x^3 term is the nonlinear spring term that will induce chaotic motion.

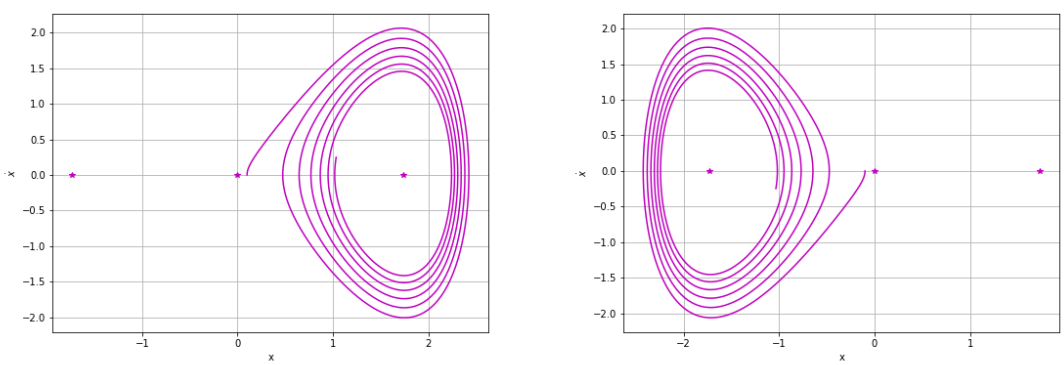
We will approach this problem in two different steps. First, we will analyze a more simple version of this system by ignoring the driving force ($F = 0$). Second, we will add the driving force and solve the equation where $F \neq 0$. We will solve the driven case at two different angular velocities (ω) to demonstrate how sensitive this system is to small changes in the initial condition.

3.1 Non-Driven System

Examining the behaviors of a more simple, non-driven system will allow us to understand complex systems more easily. Consider the case where there is no driving force ($F = 0$). In order to solve for the solution to this system, we will find the equilibrium points. We know that equilibrium points are found when the velocity ($\dot{x}$) and acceleration ($\ddot{x}$) are equal to zero. In this case, the equation of motion is simplified to

$$\begin{aligned} \alpha x + \gamma x^3 &= 0 \\ x &= 0, -\sqrt{-\frac{\alpha}{\gamma}}, -\sqrt{\frac{\alpha}{\gamma}}, \end{aligned} \tag{2}$$

where $\alpha = -3$, $\beta = 0.05$, and $\gamma = 1$. With these parameters, the values of x becomes 0, -1,73 and +1.73.



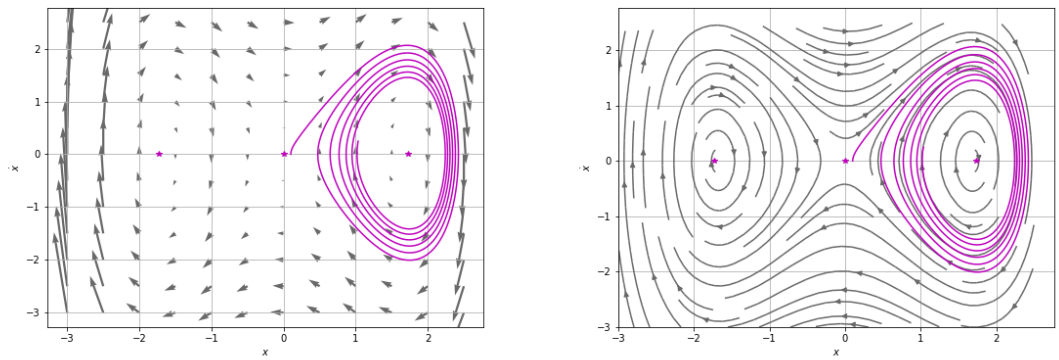
(a) x = +0.1

(b) x = -0.1

Figure 1: Phase-space diagrams of the solution to the non-driven case with initial conditions (a) $x = +0.1$ and $\dot{x} = 0$ and (b) $x = -0.1$ and $\dot{x} = 0$ in magenta. The equilibrium points are depicted as magenta stars.

We can use these parameters to numerically solve for the motion of this non-driven oscillator and evaluate the stability of these equilibrium points. We used the `odeint` numerical solver on Python to solve for the motion and generate a phase-space diagram for this oscillator (figure 1). We know that when an equilibrum point is stable, the particle will gradually move towards that point. On the other hand, all particles will move away from an unstable equilibrum point. In figure 1, we can observe one unstable point and two stable points. The point where $x = 0$ is unstable as the phase seems to be diverging from this point. The equilibrum points where $x = \pm\sqrt{-\frac{\alpha}{\gamma}}$ are both stable, as the phase seems to be circling around these points while gradually moving inward.

We can further investigate the stability of these points by generating a quiver plot (figure 2a) and a stream plot (figure 2b) to evaluate the path of an object in this system. If the vectors and the streams appear to be moving away from a point, that point is considered unstable. If the vectors and streams move closer towards a point, however, that point is considered a stable equilibrium point. Figure 2 confirms that $x = 0$ is unstable as the vectors and trajectories are pointing away from this equilibrum point, whereas the vectors and trajectories around $x = \pm\sqrt{-\frac{\alpha}{\gamma}}$ are pointing inward or towards these two equilibrum points.



(a) Quiver Plot (b) Streamplot

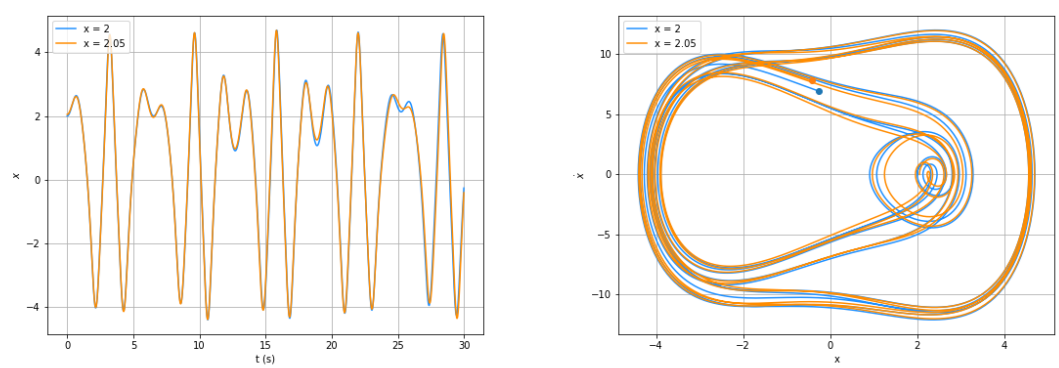
Figure 2: (a) Quiver plot and (b) stream plot plotted on top of the phase-space diagram of the solution for an object with initial conditions $x = 0.1$ and $\dot{x} = 0$. The phase-space diagram is in magenta and the quiver plot and stream plot are in light gray.

3.2 Driven System where $\omega = 2$

Now that we know more about a relatively simple system, we will add the driving force to the equation, as given in equation 1 and add two more parameters: force $(F) = 8$ and $\omega = 2$.

We solved this nonlinear differential equation using the `odeint` numerical solver on Python. In order to see how two systems evolve, we chose two different initial conditions. The first particle has an initial position (x_1) of 2 and an initial velocity $(\dot{x}_1)$ of 0, whereas the second particle has an initial position (x_2) of 2.01 and an initial velocity $(\dot{x}_2)$ of 0. These two points start off very close to one another and by numerically solving this equation and plotting it, we can see how the trajectories of these particles evolve with respect to time.

The solution to this system can be seen in figure 3, where the light blue lines represent the first particle ($x = 2$ and $\dot{x} = 0$) and the orange lines represent the second particle ($x = 2.05$ and $\dot{x} = 0$). We plotted the position as a function of time (figure 3a) to visualize how the positions evolve with time and emphasize the similarities in the trajectories of the two different particles. We also generated a phase-space diagram (figure 3b) to visualize the changes of the particles' locations relative to their velocities.



(a) Position vs. Time (b) Phase-Space Diagram

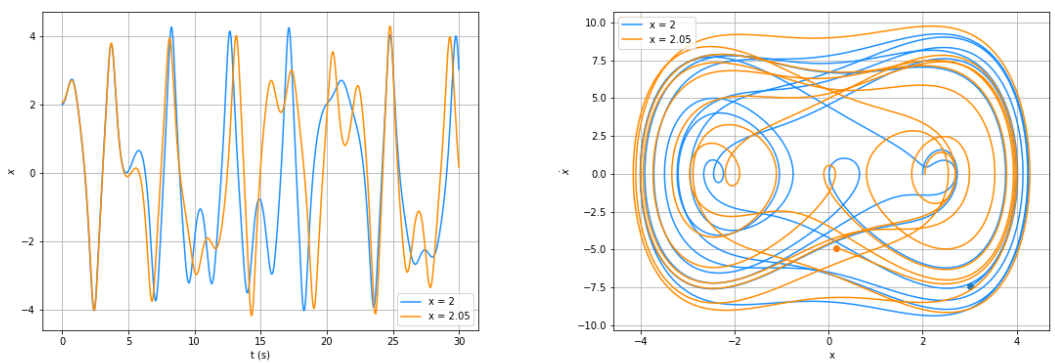
Figure 3: (a) The particles' position as a function of time and (b) phase-space diagram of the solution for a particle with initial conditions $x = 2$ and $\dot{x} = 0$ in light blue and another particle with initial conditions $x = 2.05$ and $\dot{x} = 0$ in orange.

In this condition, the two particles start with slightly different initial conditions but show similar behaviors. The position as a function of time (figure 3a) is a bit asynchronous at certain points, such as the peaks, but such differences are negligible. In addition, the phase-space of the two points do not perfectly overlap with one another and the final points are approximately 0.14 m apart, but such a small difference can be neglected. Therefore, we can conclude that this driven oscillator is not chaotic.

3.3 Driven System where $\omega = 1$

In order to observe some more interesting behavior, we solved the equation of motion with a new ω value, where $\omega = 1$, instead of 2. Using the `odeint` solver, we found the numerical solutions to this condition and plotted the solution, which can be seen in figure 4. In this figure, the light blue lines represent the first particle ($x = 2$ and $\dot{x} = 0$) and the orange lines represent the second particle ($x = 2.05$ and $\dot{x} = 0$). We plotted the position of the particles as a function of time (figure 4a) and the phase-space diagram (figure 4b) for this condition.

In figure 4a, we can see that the trajectories of the two particles start at approximately the same point and their trajectories align for the first five seconds. Then, the two trajectories start to disagree with one another and become asynchronous. The final point after thirty seconds is approximately 2.85 m away from one another. Even in the phase-space diagram in figure 4b, the trajectories barely align with one another. Therefore, when the angular velocity (ω) becomes 1, this driven oscillator shows chaotic motion.



(a) Position vs. Time

(b) Phase-Space Diagram

Figure 4: (a) The particles' position as a function of time and (b) phase-space diagram of the solution for a particle with initial conditions $x = 2$ and $\dot{x} = 0$ in light blue and another particle with initial conditions $x = 2.05$ and $\dot{x} = 0$ in orange. Here, $\omega = 1$.

3.4 Discussion

We have analyzed the motions of a driven oscillator with two different angular frequency (ω) values. Even though a relatively small change was introduced, the differences in the particles' motions over the first thirty seconds was drastic. In the case where angular frequency was equal to two, the movement of the two particles were fairly identical and non-chaotic. After decreasing the angular frequency to one, the two particles showed chaotic behaviors. After thirty seconds, we can see that the two particles were located far away from one another. These examples demonstrate this nonlinear system's sensitivity to its initial conditions.

4 The Pointcare Section

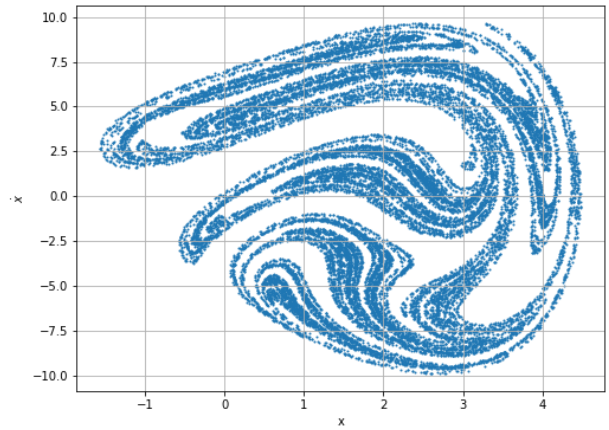


Figure 5: Poincare section for 100,000 periods.

In addition to determining the motion of the particles as a function of continuous time, we can also visualize the motion in state space at each period (one-cycle intervals) [4]. By iterating over long periods and plotting the position (x) and acceleration ($\dot{x}$) of each cycle, we can create a Poincare section that will visualize the motion of the particles over numerous cycles. The Poincare section allows us to analyze the long-term stability of the system [1] to determine if the Duffing Oscillator shows chaotic motion. The Poincare section for this Duffing Oscillator is seen in figure 5. It can be seen in this figure that the motion never repeats itself, which confirms that this Duffing Oscillator is in fact a nonlinear, chaotic system.

5 Conclusion

In this paper, we evaluated the equation of motion of a Duffing Oscillator, a second-order differential equation that is dependent on a nonlinear term, x^3. We have discussed earlier that this higher-order term is what causes chaotic motion of nonlinear systems. This factor, however, comes into effect more drastically at certain initial conditions. Through our analyses, we found that a small decrease in the angular frequency (ω) dramatically changed the behavior of this oscillator from synchronized motion to chaotic. Therefore, this system is incredibly sensitive to the initial conditions.

References

[1] Britannica. *Poincare section*. URL: `https://www.britannica.com/science/Poincare-section`.

[2] Stanford Encyclopedia of Philosophy. *Chaos*. 2008. URL: `https://plato.stanford.edu/entries/chaos/`.

[3] Stanford Encyclopedia of Philosophy. *Henri Poincare*. 2013. URL: `https://plato.stanford.edu/entries/poincare/`.

[4] John R. Taylor. *Classical Mechanics*. University Science Books, 2005.